

Closing the course — what this last hour gets you

- ✓ Capstone first— your group maps one real DOM DoIT extract job onto the GCP target architecture
- ✓ Then the fast retrace— two days against your migration map; recall, not new content
- ✓ One honest look at what changes Monday— your worksheet is backlog draft zero
- ✓ Where to go next— the 3-day Data Warehousing with BigQuery course and free practice paths
- ✓ Open questions and evaluations— the last 20 minutes are yours

The last 70 minutes

15:20–15:50

Capstone workshop — groups of 3–4 map one real extract job: source → ingest → Bronze → Silver/Gold → Power BI

WORKSHOP

15:50–16:10

Course summary — the two days retraced against your migration map, one more time, fast

16:10–16:30

Flex / open questions / evaluations — plus what comes next (the 3-day BigQuery data warehousing course)

Capstone: map one of your real extract jobs

30 minutes — 20 build + 10 share-out · Groups of 3–4 · whiteboard or paper · worksheet template in your pack

1. Pick ONE real extract job your team owns — the most typical one, not the biggest or simplest
2. Fill the template left to right: job → source → ingest → Bronze → Silver → Gold → orchestration → serving
3. Decide: one tool per box; being stuck on a box is the valuable part — flag it for share-out
4. Answer 'first thing that breaks' honestly — schema drift, PII, 10× volume, a late county file
5. Share-out: 2–2.5 min per group — read the job aloud, then walk the map left to right

Why it matters — Single-org room, so real systems are safe to name. This page is the artifact you take back to work — backlog draft zero for the migration project.

The template: one real job, left to right

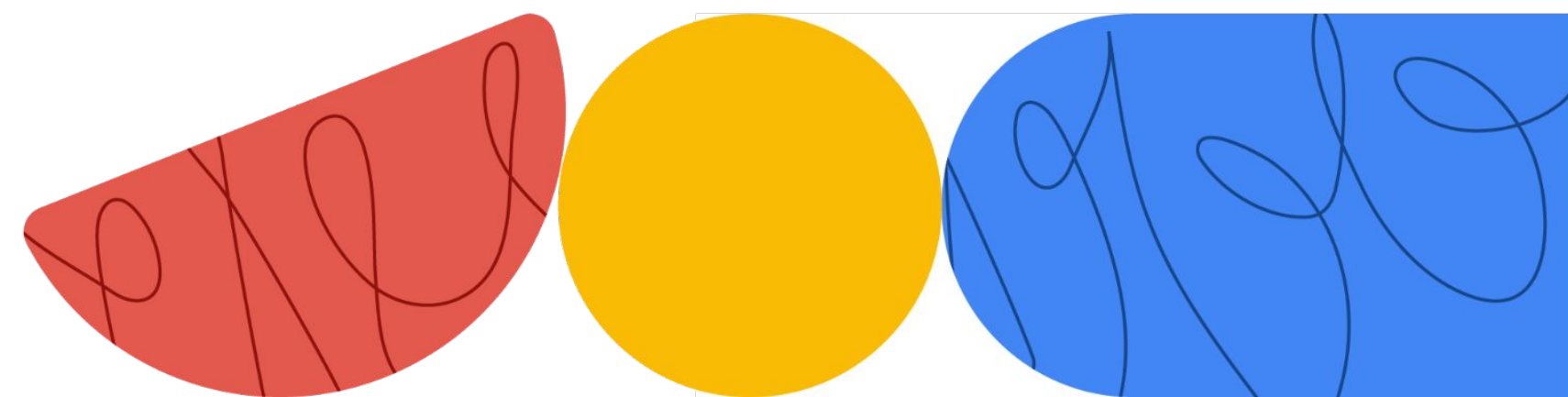
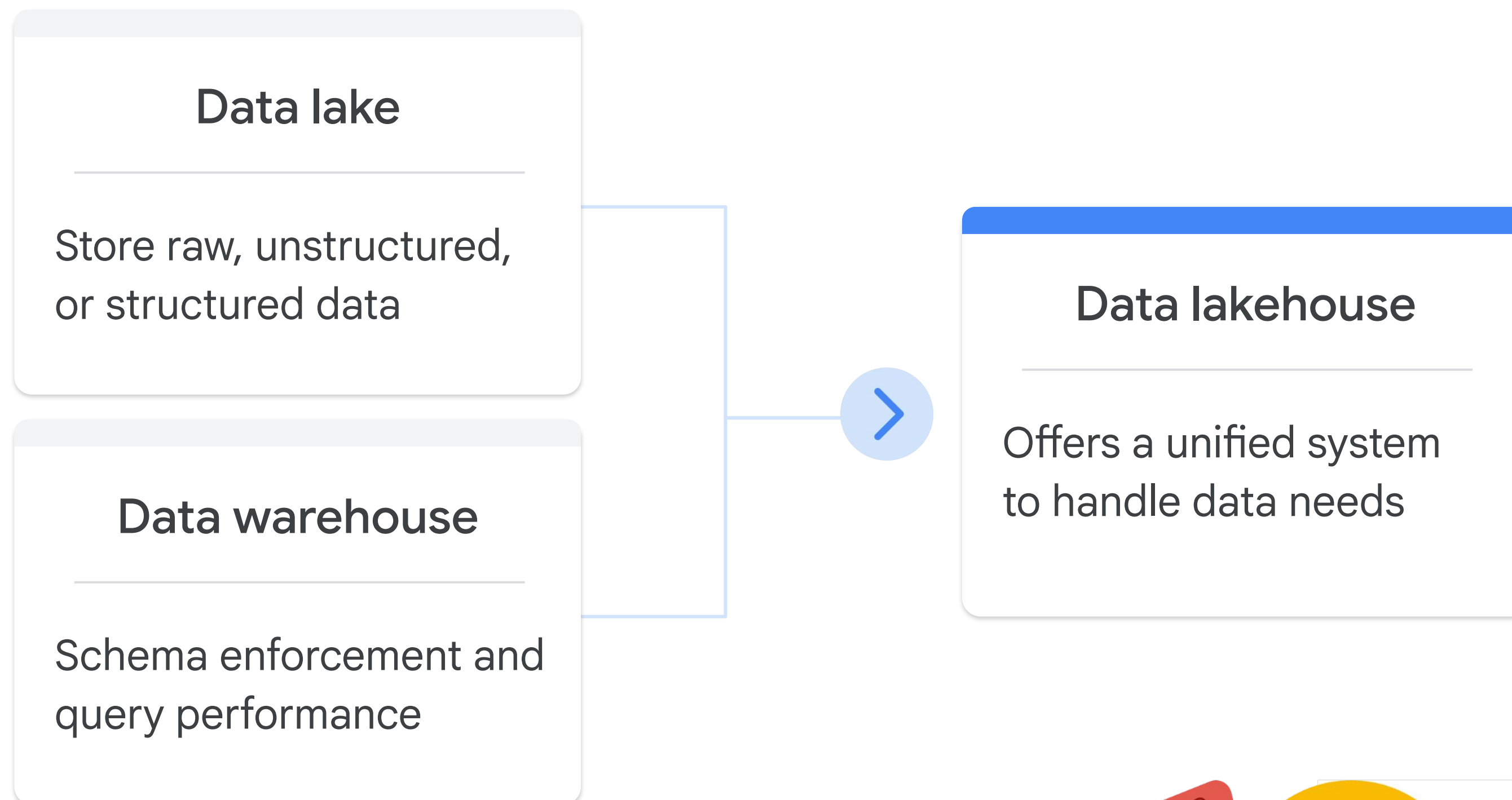
Every box has an answer from the last two days.

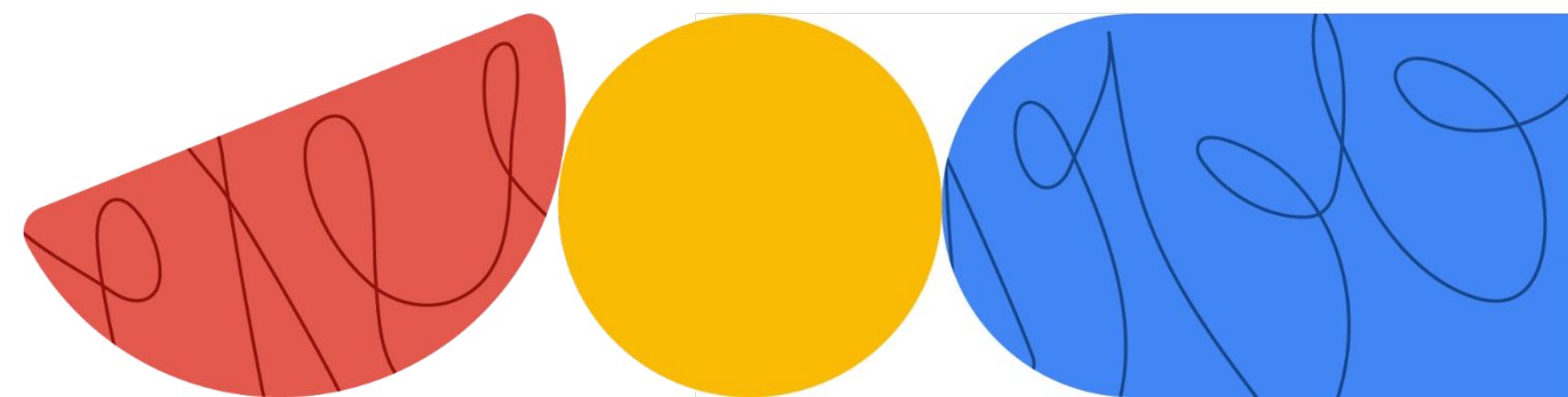
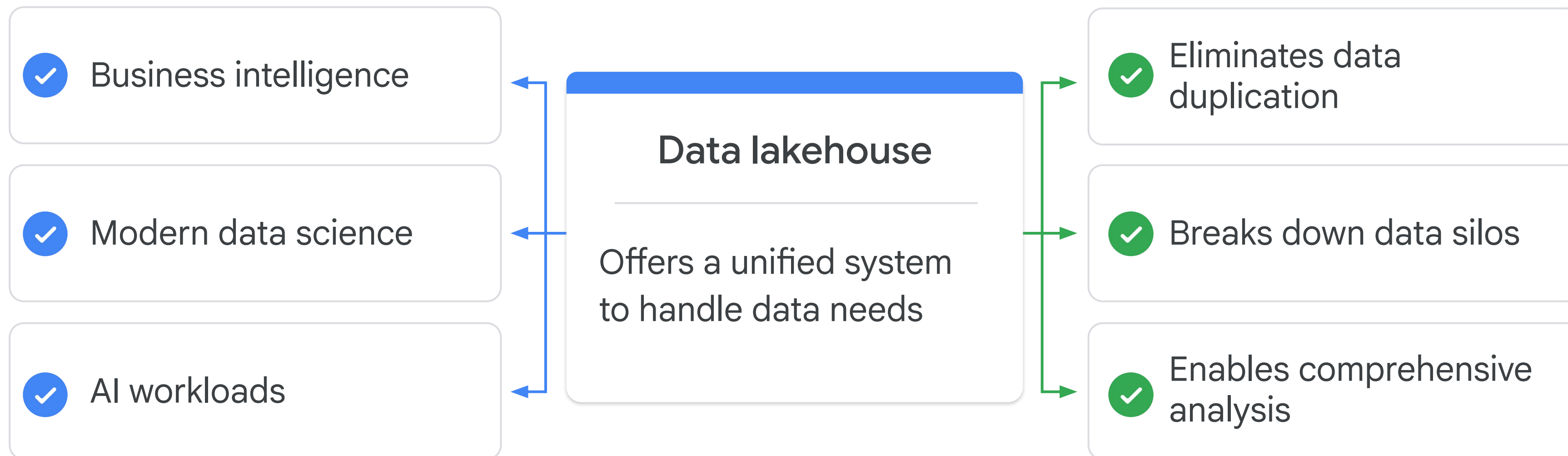
- The job— name it: one real SQL Server extract you own or babysit
- Source system— tables or files? Size? Change rate? PII?
- Ingest— Datastream CDC for databases; Storage Transfer + LOAD DATA for files
- Bronze— raw and replayable: GCS files, BigLake, or Iceberg; no cleaning here
- Silver→Gold— Dataform models dedupe and conform, then marts partitioned + clustered
- Serve— release configs + Scheduler; Power BI Import-first, refresh after the release

Stuck on ingest? Database → Datastream; files → Storage Transfer + LOAD DATA. Nightly tolerance → batch; always-current → CDC.

What a good map looks like — debrief

- Every box decided— one tool per cell; an 'or' means the decision got deferred — say so
- Bronze stays raw— it answers 'what did the source say'; Silver answers 'what does it mean'
- Dataform owns Silver→Gold, in git— transforms anywhere else need a stated reason
- Orchestration is boring— Scheduler + release configs; Composer when pipelines multiply
- Power BI Import-first— DirectQuery only with a stated freshness need
- 'First thing that breaks' is honest— those answers start your migration risk register



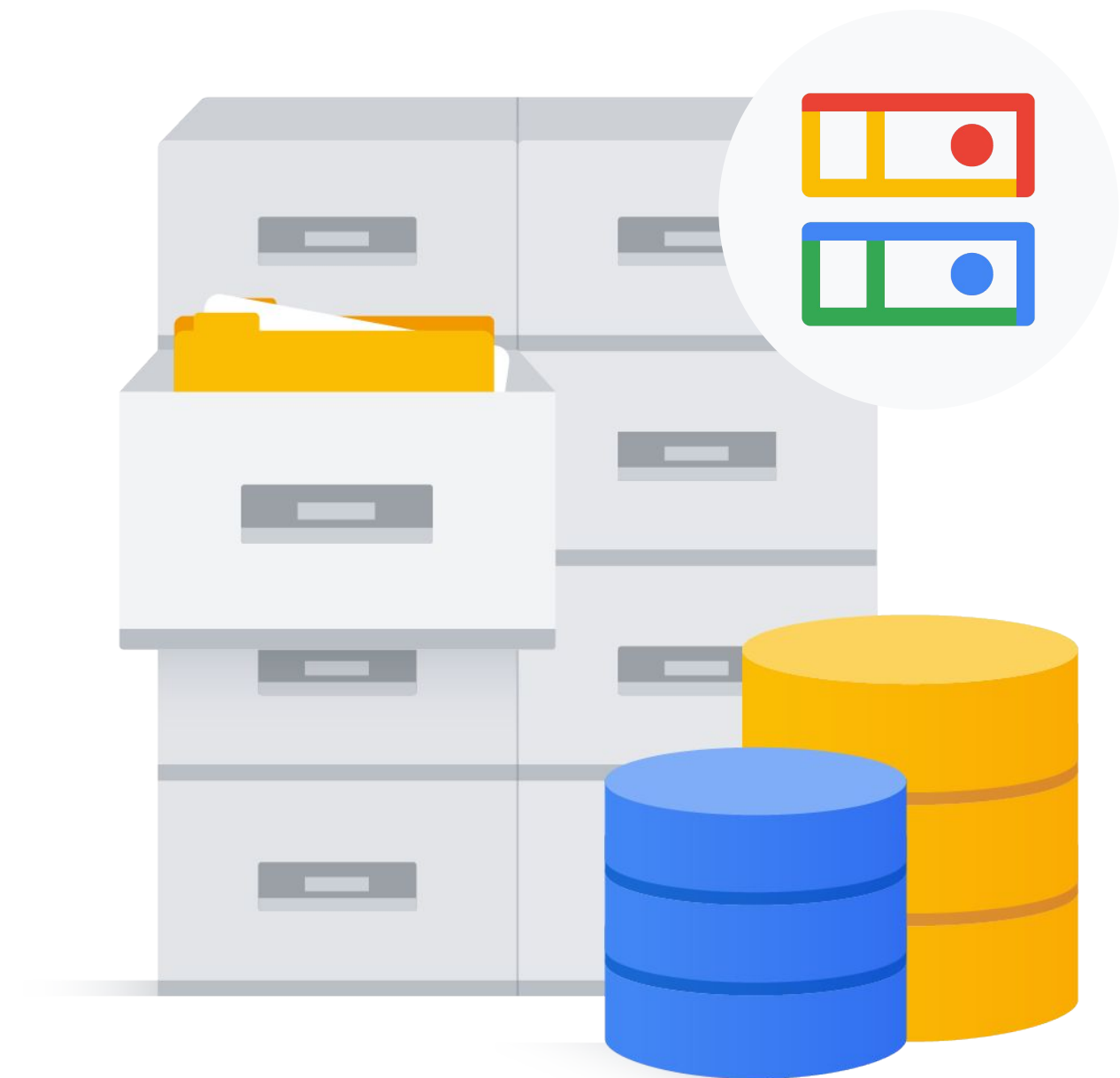


How Google Cloud services come together
to **form a modern, efficient, and scalable
data architecture.**

Cloud Storage

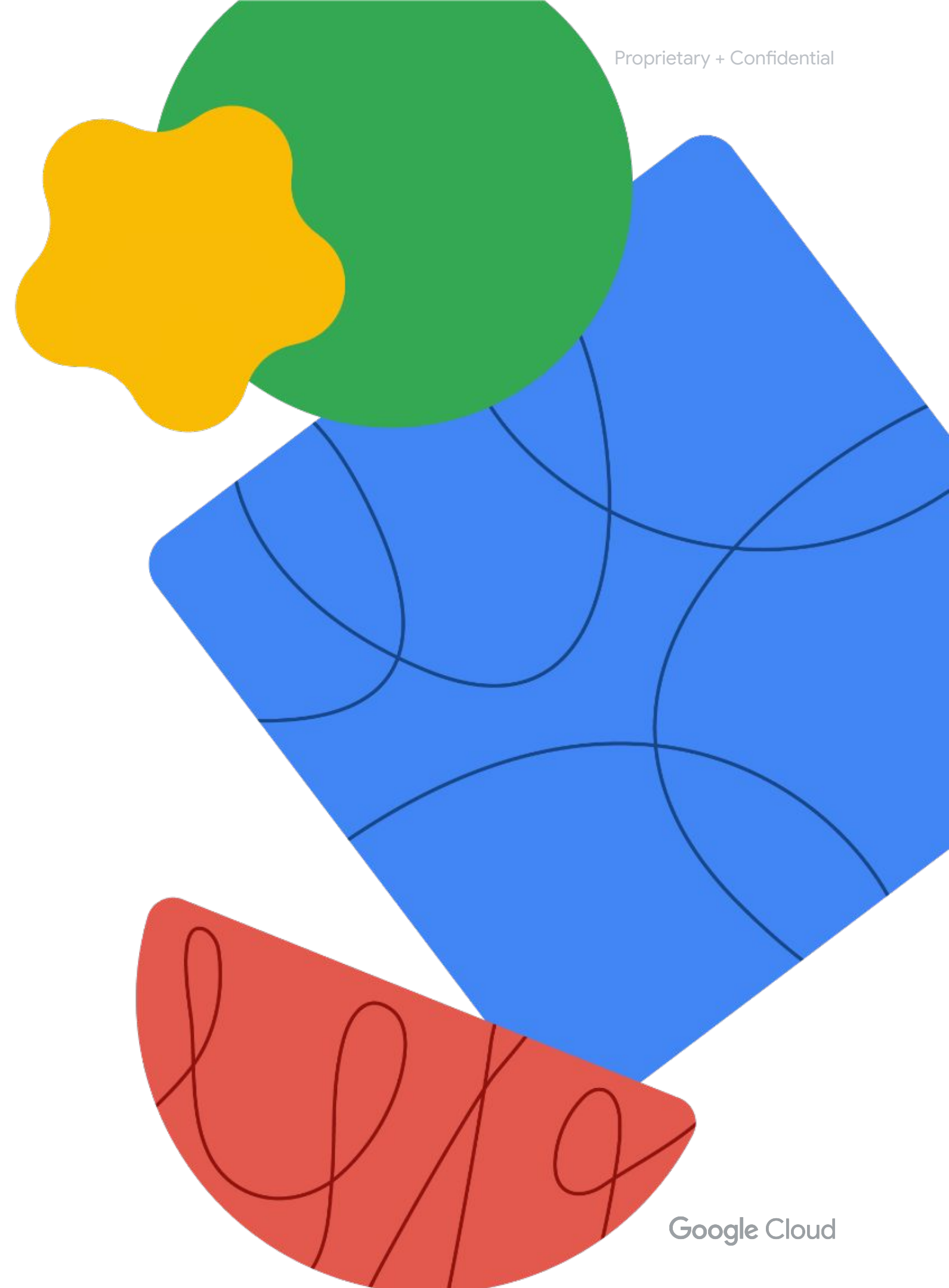
The foundational, cost-effective object storage for all raw data, including

- ✓ Structured
- ✓ Semistructured
- ✓ Unstructured
- ✓ Multimodal



Apache Iceberg

A data lakehouse needs structure and performance. Open table formats, such as Apache Iceberg, provide this.



What does Apache Iceberg provide?

01

Schema evolution

Allows schema changes (such as adding/renaming columns) without impacting existing data or applications.

02

Hidden partitioning

Automatically manages how data is partitioned for efficient querying.

03

Time travel

Enables querying past data versions for auditing and debugging, allowing users to reproduce exact data states from previous points in time.

04

Atomic transactions

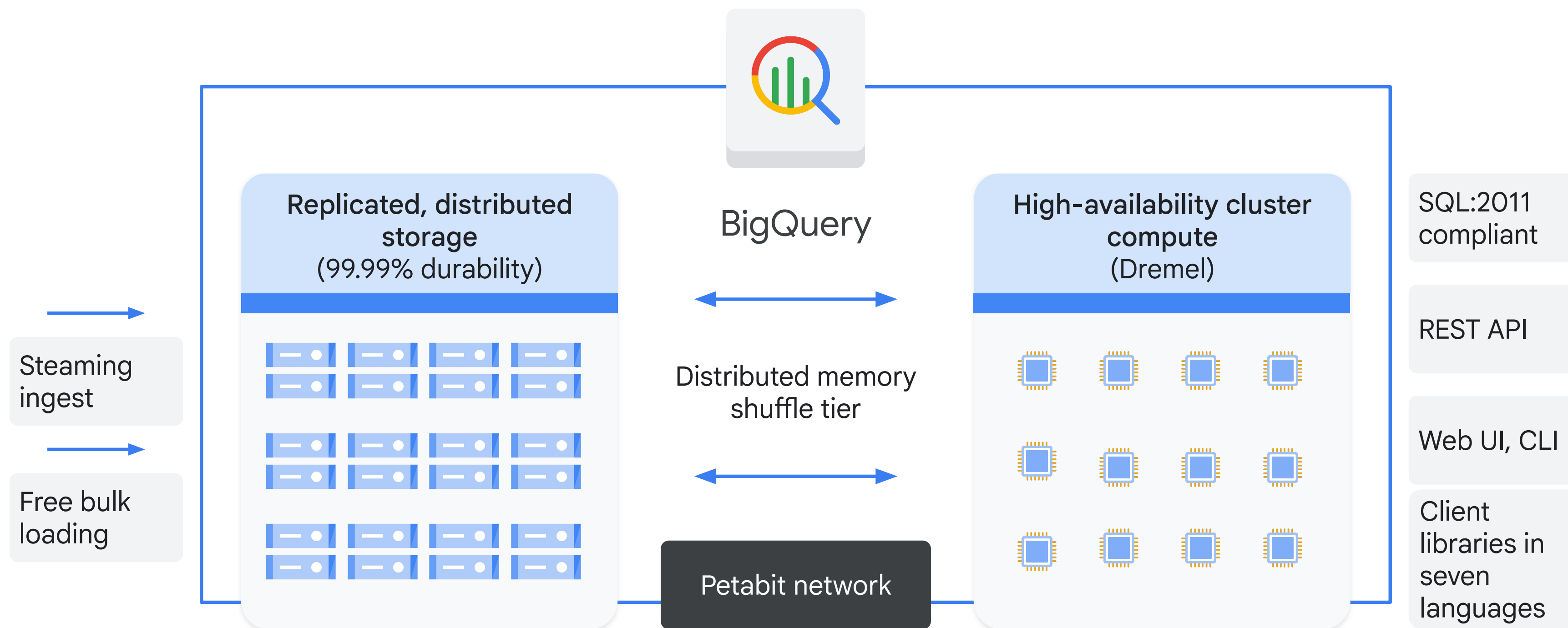
Enable reliable, concurrent data operations, preventing corruption when multiple processes read/write to the same tables.

BigQuery

Enterprise-grade, fully managed,
serverless data warehouse.



BigQuery fundamentals (1 of 2)



BigQuery fundamentals (2 of 2)

Through Lakehouse technology, BigQuery can directly query open-format data (such as Apache Iceberg tables) in Cloud Storage or other clouds, applying BigQuery's powerful optimizations, such as partitioning and clustering, by leveraging Apache Iceberg's metadata.



Lakehouse

01

Allows you to create Lakehouse tables over data in object stores (such as Cloud Storage).

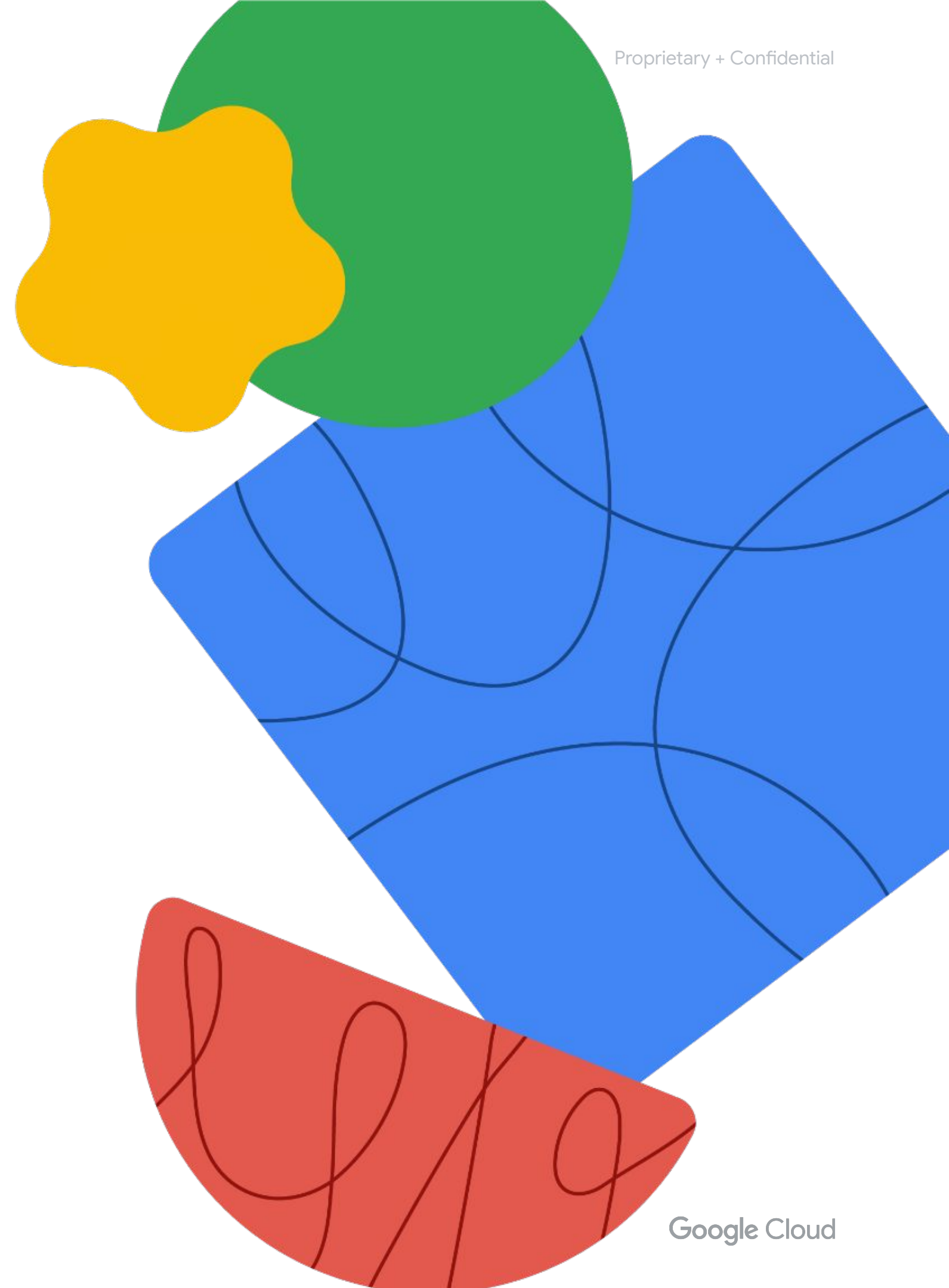
02

External tables reference data lake files, enabling unified governance and performance without duplicating data storage.



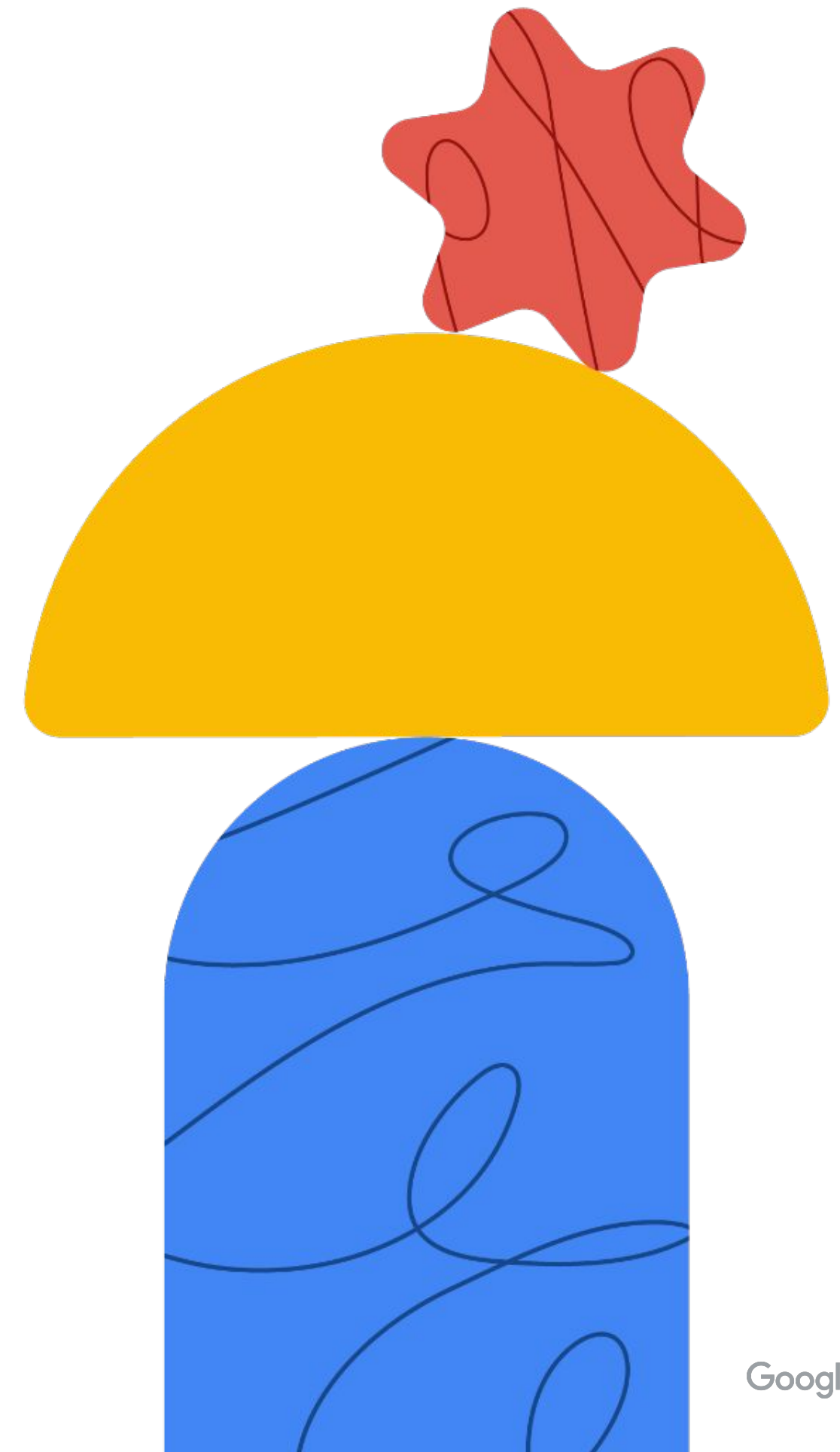
Knowledge Catalog

Unified data fabric and centralized metadata hub for discovering, managing, monitoring, and governing data across the data lakehouse.



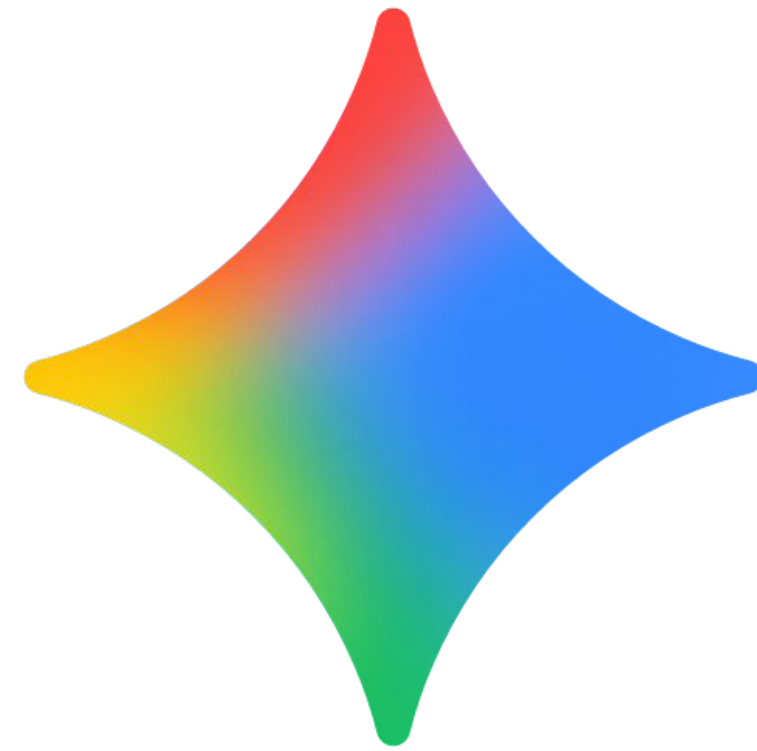
Sensitive Data Protection

Automates the discovery, classification, and protection of sensitive data using techniques, such as masking, tokenization, or redaction.



BigQuery ML and Agent Platform

Enable advanced analytics and ML
directly on data lakehouse data.



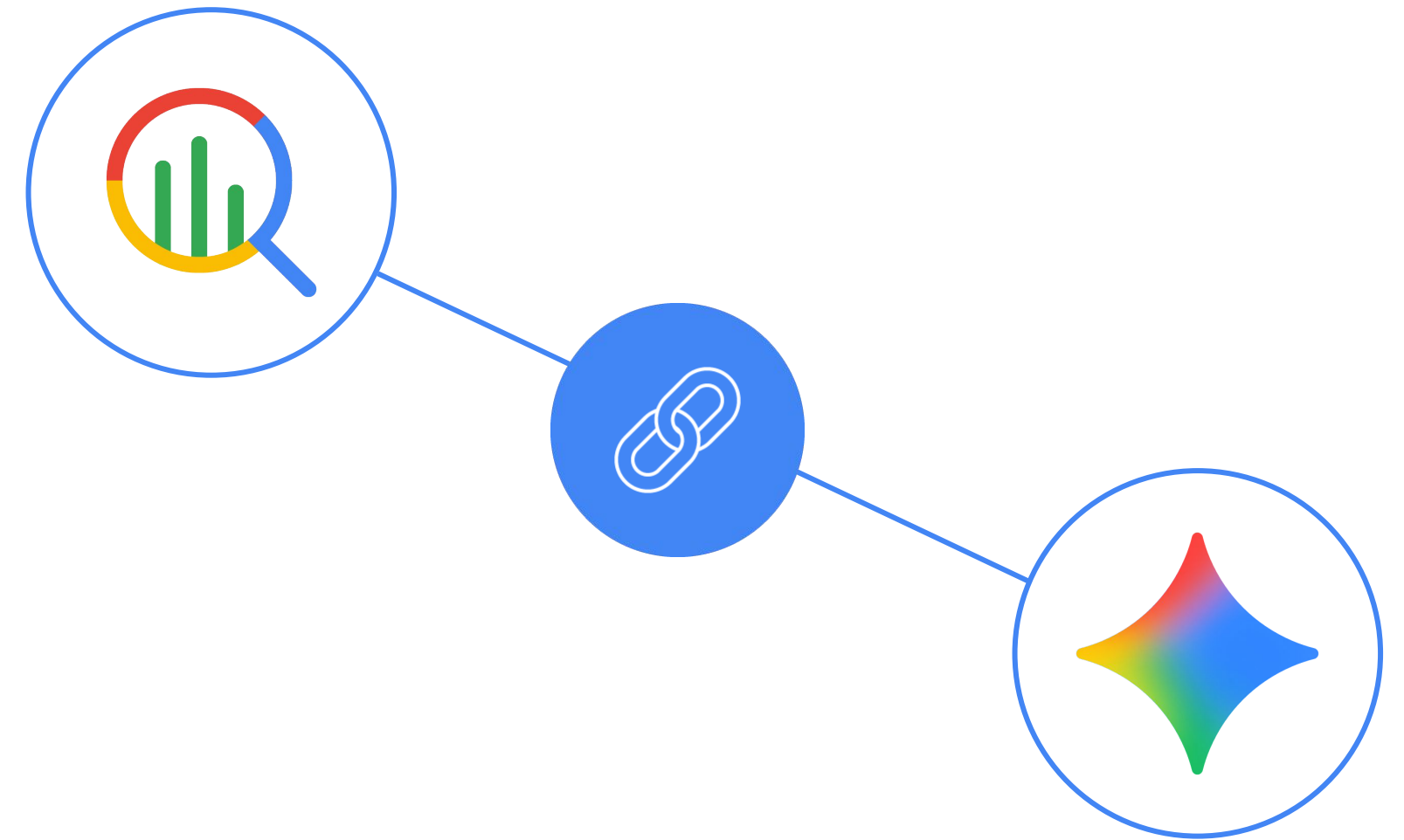
BigQuery ML and Agent Platform

01

BigQuery ML allows data analysts to build and deploy ML models using familiar SQL queries.

02

Agent Platform offers a comprehensive platform for complex custom models and MLOps, fully integrated with BigQuery.





Course summary

The two days, end to end

- ✓ Day 1 AM— console, projects, IAM, your class dataset; your first BigQuery load in a new console
- ✓ Day 1 PM— Datastream CDC killed the nightly extract; three load paths; pin schemas; CSV vs Parquet
- ✓ Day 1 close— Dataform owns Silver→Gold in git; Scheduler + release configs = SQL Agent, translated
- ✓ Day 2— lakehouse + Iceberg Bronze; governance and the medallion matched your architecture doc
- ✓ The spine— kill the extracts → land in BigQuery → Bronze/Silver/Gold → Dataform → Power BI

Where you go from here

- Data Warehousing with BigQuery— 3 days, being scheduled; goes deep on everything Day 2 previewed
- Your capstone map— photograph it tonight; it's backlog draft zero for the migration project
- Reference cards in your student pack— keep-forever desk aids, incl. the Power BI connector guide
- Practice free— 1 TiB/month query free tier, bigquery-public-data, the DIN page, dataform-demo repo
- Optional homework— the serverless Spark lab from Day 1, still waiting in your Skills classroom
- Evaluations— two minutes before you leave; they shape what we schedule for you next

Thank you for attending this training!

We love your feedback! Please take
a minute to complete the survey and
help us improve our courses.



Google Cloud